

C: Use of a spreadsheet

Consider again the fitting procedure as described in sec. 3. Once the best values of v_0, a, β have been found, it is easy to check in a spreadsheet that the least squares sum really is as small as found. Using Lotus-123, I have made such a check for all the measurements considered in the present notes. One just writes the guess for the good values, and the spreadsheet (see fig. C1) calculates the corresponding values of the least squares sum as well as the standard deviation. Behind the sheet shown in fig. C1, there are other sheets, first one for the empty merry-go-round and then one for each of the models considered. The one for the Ferrari GTO with open windows is shown in fig. C2. In that one can also see how it automatically suggest a good guess (found as described in appendix D) from which the fitting procedure can be started.

The Spreadsheet program and the FPro program confirm each other in the sense that the same values of v_0, a, β give the same least squares error when the same set of measured velocities $u_0, u_1, u_2, \dots$ is considered. The values of v_0, a, β appearing in fig. C1 are, however, not exactly the same as those obtained when the fitting is carried out by use of FPro. The reason for this is the following.

Years ago, before the existence of FPro, I made a program myself, designed to solve the specific least squares problem raised by the merry-go-round. The values of v_0, a, β appearing fig. C1 are determined by use of this program on a series of decreasing velocities $u_0, u_1, u_n, \dots$ which my students measured at that time. It seems that in almost all cases, my program has lead to a least squares sum which smaller than the one found by FPRO as described in the previous sections. The differences in the best v_0, a, β are quite small - but here and there they are large enough to give a noticeable difference in the two differences Δf and thereby in the values of the two constants in

$$C_d = \frac{c_1}{v} + c_2 \quad (C1)$$

The effect of this is seen in the C_d of the car models in fig 7.1. As one can see, it amounts to something like 1-2% . It probably requires a careful analysis to see if there is anything significant behind these differences¹. Unfortunately I cannot use the old program any more - it cannot run on todays computers, and I have not had the occasion to rewrite/update it appropriately.

¹ My wish to study these differences is the reason that the parameter v_0 in the spreadsheet shown in fig. C1 is varied in a slightly awkward way, namely by varying how much it differs from u_0 . This is how it was done in the old program.

Data for the merry-go-round			Properties of the surrounding air					
Radius in orbit		958,0 mm	Temp	20	centigrades			
Orb length at orbit (calculated elsewhere)		424,107 mm	Pressure	1007	hPa			
Length of perimeter		6,0193 m	Density	1,210	kg/m ³			
From centre to counterbalance		331,0 mm	Viscosity	1,82E-005	Pa*s			
		B	C	D	E	F	G	H
		no model	Sphere	Circular disk	Mercedes SSK 1928	Ford fireengine	Ferrari GTO Open wind	Ferrari GTO Closed wind
v_0	m/s	9,1074	7,396	8,734	6,948	5,654	6,873	6,590
$(v_0-u_0)*1000 \rightarrow$	m/s	4,4000	-0,6987	0,3935	3,5422	-1,4553	-0,0509	-0,8613
$d\alpha*1000 \rightarrow$	1/s	2,6520	2,1059	2,4264	1,4839	1,2241	1,3158	1,4541
$\beta*1000 \rightarrow$	1/m	2,3670	3,6734	8,2106	3,9985	3,2038	2,2498	2,1765
Least Square Sum*1000	(m/s) ²	0,47	0,72	0,28	1,40	0,47	1,04	0,52
Standard deviation	mm/s	3,2	4,0	3,3	6,9	3,9	4,7	3,6
$w = d\alpha/\beta$	m/s	1,120	0,573	0,296	0,371	0,382	0,585	0,668
Equivalent mass m	g	560	598	570	1113	1251	1310	1310
$f1 = m*d\alpha$	mN/(m/s)	1,485	1,259	1,383	1,652	1,532	1,724	1,905
$f2 = m*\beta$	mN/(m/s) ²	1,326	2,197	4,680	4,452	4,009	2,948	2,852
Frontal area A	cm ²	---	30,68	45,4	60,8	71,9	60,2	60,2
$\Delta f1$	mN/(m/s)	---	-0,226	-0,102	0,167	0,047	0,239	0,420
$\Delta f2$	mN/(m/s) ²	---	0,871	3,355	3,126	2,684	1,623	1,527
c1	m/s		-0,122	-0,037	0,045	0,011	0,066	0,115
c2	dim. less		0,469	1,221	0,850	0,617	0,445	0,419

Figure C1: Spreadsheet by which one can determine the least square sum corresponding to a guess for the values of v_0, α, β . The arrows indicate where these values should be entered. The value of v_0 is entered by the amount it deviates from u_0 - the reason for this slightly awkward procedure is explained in the footnote on p. C-1.

Model:		Ferrari GTO 1984				
Notes:		The model is driving with its windows closed - that is, covered by tape.				
Data for merry-go-round and model				Good start for fit:		
Frontal Area --->		60,2 cm ²		v0 = u0		6,596
Mass me. Without model --->		557 g		If 19 < last n <30:		
Mass of model --->		634 g		alfa*1000		1,591
Counterbalance--->		1000 g		beta*1000		2,164
Mass m. With model		1310 g		If 29 < last n <40:		
				alfa*1000		1,099
				beta*1000		2,247
Measurements on velocities				If 39 < last n :		
Least Square Sum * 1000		0,52 (m/s) ²		alfa*1000		1,248
Standard Deviation		3,6 mm/s		beta*1000		2,225
n	tn *100	tn	un	vn	vn-un	(vn-un) ² *1000
	ms	ms	m/s	m/s	mm/s	(m/s) ²
0	6430	64,30	6,596	6,590	-5,7	0,03
1	6530	65,30	6,495	6,496	0,9	0,00
2	6620	66,20	6,406	6,402	-4,1	0,02
3	6710	67,10	6,321	6,310	-10,2	0,10
4	6820	68,20	6,219	6,220	0,9	0,00
5	6920	69,20	6,129	6,130	1,2	0,00
6	7020	70,20	6,041	6,041	-0,0	0,00
7	7120	71,20	5,957	5,954	-2,5	0,01
8	7230	72,30	5,866	5,868	1,9	0,00
9	7340	73,40	5,778	5,783	4,8	0,02
10	7440	74,40	5,700	5,699	-1,5	0,00
11	7550	75,50	5,617	5,616	-1,3	0,00
12	7670	76,70	5,529	5,534	4,8	0,02
13	7780	77,80	5,451	5,453	2,2	0,00
14	7900	79,00	5,368	5,374	5,3	0,03
15	8010	80,10	5,295	5,295	0,4	0,00
16	8140	81,40	5,210	5,218	7,4	0,05
17	8260	82,60	5,134	5,141	6,5	0,04
18	8370	83,70	5,067	5,065	-1,7	0,00
19	8500	85,00	4,989	4,991	1,2	0,00
20	8630	86,30	4,914	4,917	2,7	0,01
21	8760	87,60	4,841	4,844	2,9	0,01
22	8890	88,90	4,771	4,773	2,0	0,00
23	9030	90,30	4,697	4,702	5,1	0,03
24	9160	91,60	4,630	4,632	1,9	0,00
25	9290	92,90	4,565	4,563	-2,3	0,01
26	9430	94,30	4,497	4,495	-2,6	0,01
27	9580	95,80	4,427	4,428	0,6	0,00
28	9720	97,20	4,363	4,361	-1,9	0,00
29	9860	98,60	4,301	4,296	-5,4	0,03
30	10020	100,20	4,233	4,231	-1,4	0,00
31	10170	101,70	4,170	4,167	-2,7	0,01
32	10330	103,30	4,106	4,105	-1,1	0,00
33	10480	104,80	4,047	4,042	-4,4	0,02
34	10640	106,40	3,986	3,981	-4,9	0,02
35	10820	108,20	3,920	3,921	0,9	0,00
36	10980	109,80	3,863	3,861	-1,7	0,00
37	11160	111,60	3,800	3,802	1,7	0,00
38	11320	113,20	3,747	3,744	-2,8	0,01
39	11510	115,10	3,685	3,686	1,6	0,00
40	11690	116,90	3,628	3,630	1,7	0,00

Figure C2: Results for Ferrari GTO with closed windows. The measured velocities $u_0, u_1, u_2, \dots$ compared with the velocities $v_0, v_1, v_2, \dots$ found in the fit. As one can see, the agreement is very good. Comparing with the analogue results in figs. 3.3 and 3.3 one notes that all the times t_n have zero as their last digit. The students who made the experiment had only adjusted the clock to measure with an accuracy of 0.1 ms - but as one can see from the small least squares sum, this has been good enough. Hence there is probably not much to gain by improving the accuracy of the time measurements.

Good start for fit

Once the velocities $u_0, u_1, u_2, \dots$ are known, a good set of values of v_0, a, β whereby to start the fitting procedure, can be found by the formulas (D4). This is done in the little separate table in the upper right corner. Here v_0, a and β are measured in the same units as in fig. C1, that is m/s, 1/s and 1/m, respectively.